Question ID: c447367f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the *xy*-plane has its center at $(19, 18)$ and has a radius of $9k$. Which equation represents this circle?

Answer

A. $(x-19)^2+(y-18)^2=81k$

B. $(x-19)^2+(y-18)^2=81k^2$

C. $(x-19)^2+(y-18)^2=9k$

D. $(x-19)^2+(y-18)^2=9k^2$

Correct Answer: B

Rationale

Choice B is correct. The equation of a circle in the *xy*-plane can be written as $(x-h)^2+(y-k)^2=r^2$, where the center of the circle is (h, k) and the radius of the circle is r. It's given that this circle has a center at $(19, 18)$ and a radius of $9k$. Substituting 19 for h, 18 for k, and $9k$ for r in $(x-h)^2+(y-k)^2=r^2$ yields $(x-19)^2+(y-18)^2=(9k)^2$, or $(x-19)^2+(y-18)^2=81k^2$. Therefore, the equation that represents this circle is $(x-19)^2+(y-18)^2=81k^2$.

Choice A is incorrect. This equation represents a circle with radius $9\sqrt{k}$, not $9k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{9k}$, not $9k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{9}k$, not $9k$.